

Practical-12

Aim:

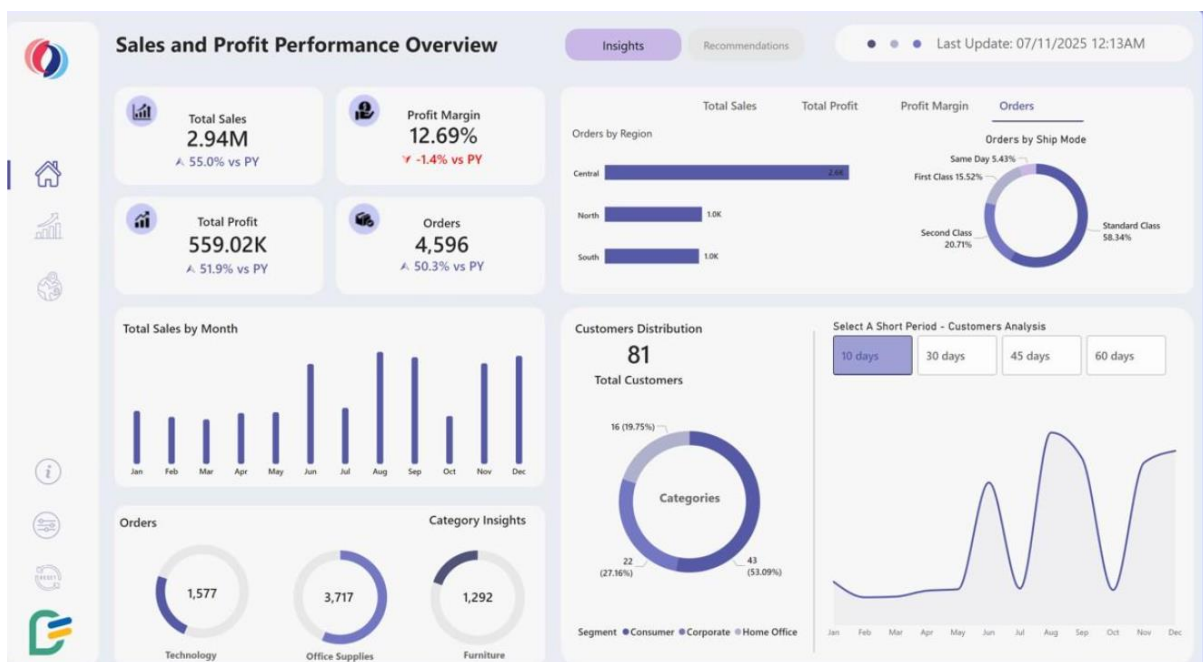
To design and develop an interactive sales dashboard that enables management to monitor sales performance across products, regions, and customer segments and identify important business trends for effective decision-making.

Problem Definition:

You are working as a Data Visualization Analyst at a retail analytics company. Management wants to monitor sales performance across products, regions, and customer segments through an interactive dashboard. Static reports are insufficient for decision-making. Your task is to design an interactive dashboard that allows management to explore sales data and identify key business trends.

Objectives/Outputs:

Tasks: Import sales dataset into Tableau / Power BI.



Clean and validate the dataset.

es("#Removed Duplicates",{{"Visit_Date", type date}})

Visit_Date

Copy

Remove

Remove Other Columns

Duplicate Column

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Group By...

Fill

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

Drill Down

Add as New Query

Decimal Number

Fixed decimal number

Whole Number

Percentage

Date/Time

Date

Time

Date/Time/Timezone

Duration

Text

True/False

Binary

Using Locale...

Query Settings

PROPERTIES

Name

Visits

All Properties

APPLIED STEPS

Source

Navigation

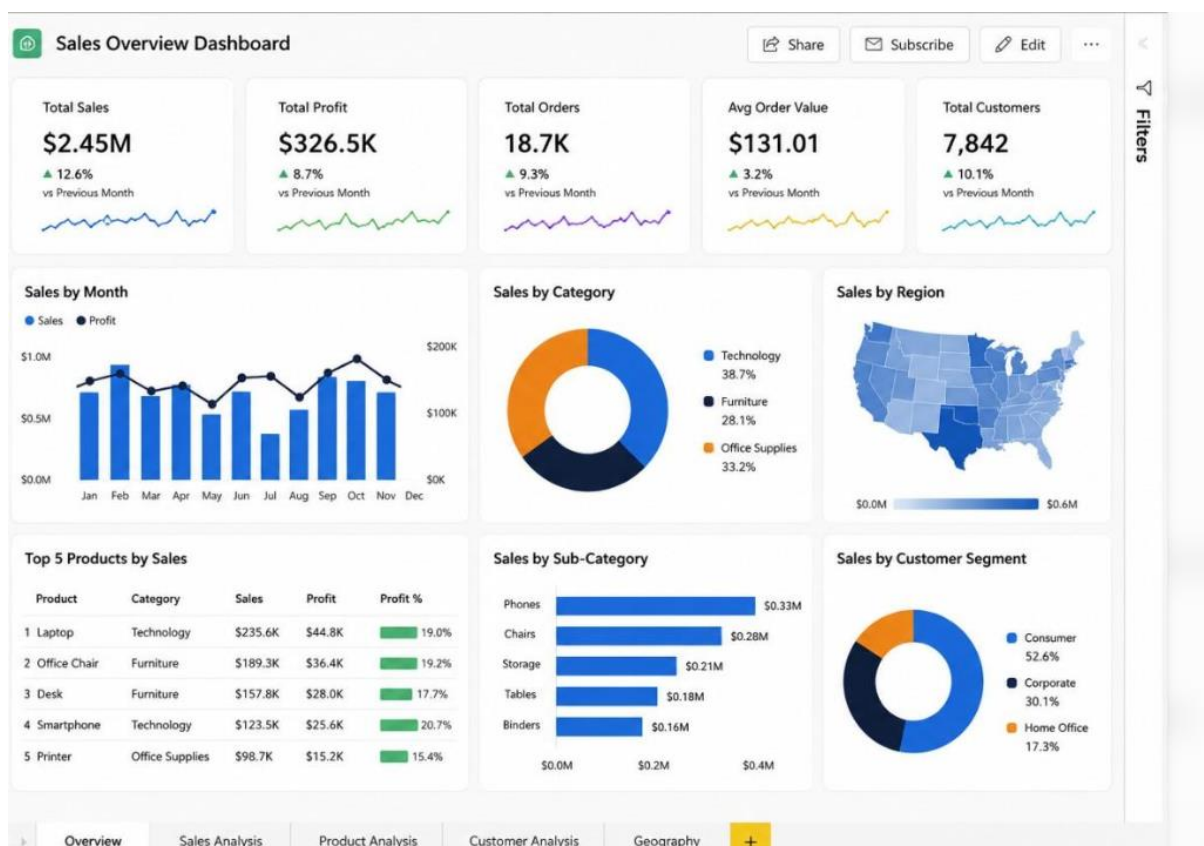
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Create KPIs for Total Sales, Profit, and Orders.



Order Date Slicer

03-01-2014
30-12-2017

Region

Select all
Central
East
South
West

Category

All

Total Sales by Category and Segment

Segment	Category	Sum of Sales (K)
Furniture	Consumer	305
	Corporate	180
	Home Office	100
Office Supplies	Consumer	285
	Corporate	180
	Home Office	105
Technology	Consumer	320
	Corporate	205
	Home Office	160

Total Orders Shipped by Modes

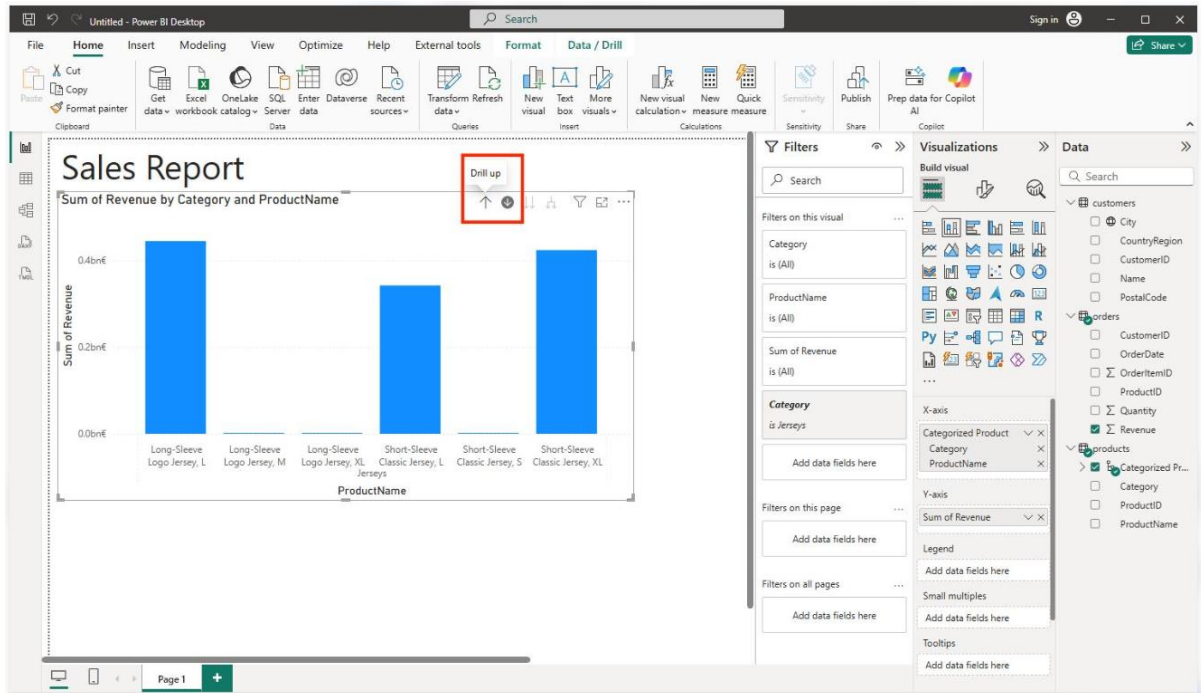
Ship Mode	Percentage	Percentage of Total
Standard Class	4.79%	54.01%
First Class	1.37%	15.56%
Second Class	1.23%	14.29%
Same Day	0.43%	5.14%

Total Number of Products Transacted Per Region

Region	Count	Percentage of Total
West	2.61K	12.58%
East	2.22K	10.89%
Central	1.82K	8.79%
South	1.31K	6.38%

The dashboard provides a comprehensive overview of the company's performance. Key metrics include 104 total stores, \$22M in sales for this year, and a target for new stores. The 'This Year's Sales by Store Number Name' chart highlights the top-performing stores, with '13 - Charleston Fas...' leading. The 'Stores Opened This Year' chart shows a steady increase in new store openings throughout the year. The 'Sales Per Sq Ft' chart identifies high-performing locations like 'Chicago' and 'New York'. The 'Total Sales Variance % by District' chart shows a significant variance in the 'FD - 01' district.

Develop drill-down functionality.



Add dashboard interactions.



Create summary cards. Publish dashboard. Interpret business insights.



Objectives

1. To analyze sales data across different products, regions, and customer segments.
2. To visualize key performance indicators (KPIs), such as total sales, profit, quantity sold, and sales growth.
3. To create interactive charts, graphs, and filters for exploring sales performance.
4. To compare product-wise and region-wise sales performance.
5. To identify top-selling products, profitable regions, and customer purchasing patterns.
6. To enable dynamic data filtering by date, region, product category, and customer segment.
7. To support data-driven business decisions through clear and meaningful visualizations.

Key Questions /Analysis:

Q1. What is the role of interactive dashboards in business analytics?

Interactive dashboards help businesses monitor and analyze data in a visual and user-friendly way. They display important information such as sales, profit, product performance, and regional trends, helping management identify patterns and make better decisions.

Q2. How do filters improve visual exploration?

Filters allow users to view specific data based on criteria such as date, region, product category, and customer segment. They make it easier to compare results, focus on relevant information, and identify trends without analyzing the entire dataset.

Q3. What are KPI indicators?

KPI stands for Key Performance Indicator. KPIs are measurable values used to evaluate how effectively a business is achieving its objectives. Examples include:

- Total sales revenue
- Total profit
- Number of units sold
- Sales growth percentage
- Customer retention rate

Q4. Why are dashboards preferred over static reports?

Dashboards are preferred because they provide interactive, real-time or regularly updated visual information, while static reports present fixed information. Dashboards allow users to apply filters, compare performance, explore trends, and identify problems more quickly.

Q5. How does user-driven exploration support decision-making?

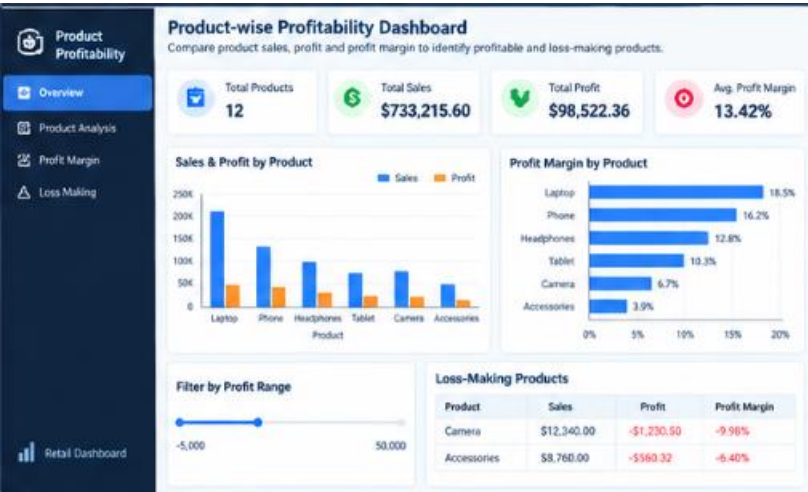
User-driven exploration allows management to interact with data according to their needs. Managers can filter sales by region, compare products, and analyze customer segments to discover opportunities or problems. This supports faster, more informed, and data-driven business decisions.

Supplementary Problems:

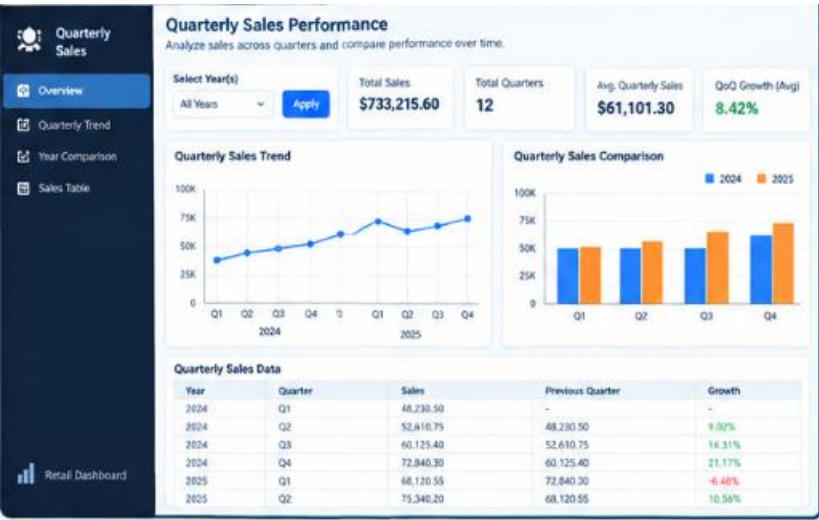
Add customer segmentation analysis.



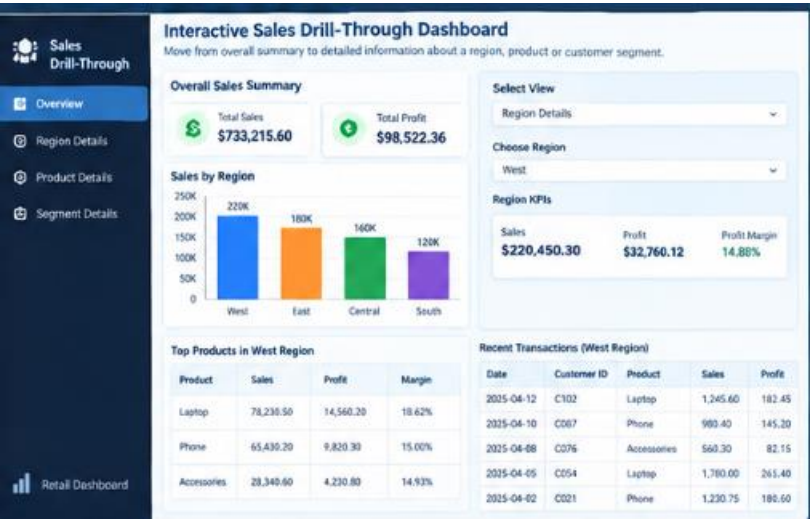
Create product-wise profitability dashboard.



Compare quarterly sales performance.



Implement drill-through pages.



Create executive summary dashboard.



Tools and Technologies Used

Tool/Technology	Purpose
Microsoft Excel / CSV	Storing and managing sales data
Python	Data processing and analysis
Pandas	Data cleaning and manipulation
Matplotlib / Seaborn / Plotly	Creating charts and interactive visualizations
Streamlit	Developing the interactive dashboard
Jupyter Notebook / VS Code	Writing and executing code

Conclusion

The interactive sales dashboard provides a comprehensive view of retail sales performance across products, regions, and customer segments. It enables management to explore data using filters, compare business performance, identify sales trends, and recognize profitable areas. Thus, the dashboard

improves data accessibility and supports faster, more informed, data-driven business decisions.